Cloakroom: A Dark Pool for Shielded Zcash with Scriptless Atomic Settlement

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Abstract. Every route out of shielded Zcash today passes through a transparent address held by an exchange, a bridge or an intent network, which publishes amounts and timing and can hold funds. Cloakroom trades shielded ZEC against tokens on an EVM chain without either. The two parties split the spend authorizing key of a fresh Ironwood address into additive shares, and the counterparty's tokens sit in an escrow whose two exits each publish one share. We give a timing model with bounded inclusion delay, finality lag and a budget of reorganizations, and prove three things. An honest taker loses at most a small premium and is never held hostage while she stays online through her claim. An honest maker never loses funds if he comes online once in a stated window, which any protocol that pays the taker's unilateral exit on the EVM chain without observing Zcash must require. A passive observer learns nothing that links the two chains beyond timing and the wallet fingerprint the transactions share. Model checking the protocol in TLA$^+$ found that a refund which is included but not yet final can be reorganized into a theft, and, at a second parameter point, was consistent with the budget every share-revealing transaction needs, which a reviewer of an earlier version derived; our checked points all have $D = F$, so the budget's form rests on its proof. Measuring 30 days of Zcash mainnet shows that timing is the binding privacy constraint. A swap transaction that is not delayed hides among almost no others. A natural count-gated delay lets an observer who knows the policy guess the transaction with probability 0.18 in the median case and above 0.85 in one case in ten; the delay our wallet uses keeps it near 1.4% per delayed transaction (median, for an observer who samples its posterior) at an average of 2.6 hours per delayed step. A swap is linked if any one of its transactions is, and its funding is anchored on both sides, so the per-swap probability is higher; we do not bound it. The implementation passes eighteen two-party scenarios, including injected reorganizations, an impostor maker, malicious quotes and a funding mined after the maker's refund. On Zcash testnet and Base Sepolia the final escrow has settled three swaps, the last with the final code, and every Zcash transaction was fully shielded; a testnet run also exposed a fund-level bug that local chains had not, now fixed.

1. Introduction

Zcash [1, 2] offers two kinds of value. Transparent value is public, as in Bitcoin [3]. Shielded value is held as encrypted notes, and a fully shielded transaction shows an observer only that it happened and what fee it paid. Most shielded ZEC is now in the Ironwood pool introduced by NU6.3 in July 2026 [4].

Shielded value stays private only while it stays shielded. Each crossing between transparent and shielded value publishes an amount and a time, and matching crossings has linked a large share of shielded activity to transparent addresses [17, 18, 19]; similar methods recently linked 17.65% of Railgun withdrawals on Ethereum [20]. On mainnet, 51% of Ironwood transactions in the last 30 days touched a transparent address (Section 8.2). Exchange is where this happens: most exchanges withdraw ZEC only to transparent addresses; NEAR Intents supports ZEC on transparent addresses only [21]; and Maya and THORChain hold ZEC in transparent vaults. Each route also adds a custodian. A user reported a $589,000 payout from a ZEC swap held in a compliance review for more than fifty days [22], and on October 1, 2026 NEAR Intents was exploited for about $3.8 million before pausing and, according to reports, recovering the funds [23].

Hash-locked atomic swaps [24, 25] remove the custodian but need scripts, which Zcash's shielded pools lack; on transparent outputs, the shared hash links both legs [26]. Monero, which also lacks scripts, swaps by locking its leg to a key split between the parties [27, 29]. Cloakroom brings this approach to Orchard-protocol keys, which the Ironwood pool uses, and adds what a deployable venue needs. We make five contributions.

- **A protocol.** A swap between shielded ZEC and any EVM token whose Zcash leg is ordinary fully shielded transactions. Claims need Ready and are committed before they are revealed, payments are pulled, and a punish window removes the hostage case (Section 4).
- **A model and proofs.** A timing model with inclusion delay D, finality lag F and R reorganizations, giving every share-revealing transaction the budget $(R + 1)D + RF$. Proofs of taker safety without hostage, of maker safety under a liveness condition, and that the condition is needed whenever the taker's unilateral exit is paid

on the EVM chain, which does not observe Zcash (Sections 3 and 5).

- **Mechanized verification.** A TLA$^+$ model checked by TLC at three small parameter points, $(D, F, R) \in \{(1,1,1),(2,2,1),(2,2,2)\}$, in which removing any of the rules R2c–R3l or L_M, or shortening either budget by one step, yields a concrete attack. Symbolic execution, mutation testing, static analysis and stateful fuzzing of the escrow (Section 6).
- **A measured privacy analysis.** A simulation argument for passive observers, anonymity sets measured on 30 days of Zcash mainnet, and the posterior of an observer who knows the delay policy, which determines the policy our wallet uses (Section 8).
- **An implementation.** Run in eighteen two-party scenarios, two of them with injected reorganizations, one with an impostor maker, two with malicious quotes and one with a late funding, and on public test networks (Section 9).

We use "dark pool" in its trading sense: a venue without pre-trade transparency. Requests and quotes in Cloakroom travel encrypted, and only the executed price becomes public. Unlike an exchange-run dark pool, Cloakroom has no central matching; it is bilateral request-for-quote between a taker and a market maker. The taker is anonymous; the maker's EVM address is public.

2. Background

Orchard keys. Orchard [2, §4.2.3] works over Pallas, $E_p : y^2 = x^3 + 5$, with prime group order q. A spending key yields a spend authorizing key $\mathsf{ask} \in \mathbb{F}_q^*$, a nullifier key $\mathsf{nk} \in \mathbb{F}_p$ and a randomness $\mathsf{rivk} \in \mathbb{F}_q$. The spend validating key is $\mathsf{ak} = [\mathsf{ask}]\mathcal{G}$, where $\mathcal{G}$ is the RedPallas spend-authorization base point. A full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{rivk})$ encodes ak by its x-coordinate [2, §5.6.4.4], so key generation fixes $\tilde{y}(\mathsf{ak}) = 0$, negating ask if needed. From the full viewing key derive the incoming viewing key $\mathsf{ivk} = \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{rivk}}(\mathsf{Extract}(\mathsf{ak}), \mathsf{nk}) \notin \{0, \bot\}$, an outgoing viewing key ovk, and mutually unlinkable diversified addresses.

Spending. A transaction is a set of actions, each pairing a spend and an output. To spend, one picks a fresh α, publishes $\mathsf{rk} = \mathsf{ak} + [\alpha]\mathcal{G}$, and signs the ZIP 244 sighash [8] with RedPallas under $\mathsf{ask} + \alpha$. A Halo 2 proof [12] shows that rk randomizes the ak of the spent note's address, that the note is in the commitment tree, and that the nullifier is correct. An action publishes a value commitment, nullifier, rk, note commitment, ephemeral key and ciphertexts.

Ironwood and NU7. A soundness flaw in the Orchard action circuit was disclosed in May 2026. NU6.2 (June 2026) fixed the circuit and re-enabled Orchard. NU6.3 (July 28, 2026) then introduced the Ironwood pool, with its own note-commitment tree, nullifier set and value pool, and limited the Orchard pool to withdrawals through a turnstile [4]. Ironwood keeps Orchard's keys and addresses, and one incoming viewing key scans both pools [5]. NU7 (testnet since October 4, 2026) shortens blocks to 25 s and keeps the transaction format [7]. A partially created transaction (PCZT) [9] separates construction, proving and signing, so that a spend can be signed by a key that no wallet holds; Zcash's FROST support uses the same interface [13, 10].

The EVM chain. We use Base, an Ethereum rollup. Its transactions are ordered by a sequencer; a transaction deposited through the `OptimismPortal` contract on Ethereum must be included within the sequencing window, 3,600 Ethereum blocks or about 12 hours, which still holds after Base's May 2026 Azul upgrade [38, 39]. We measured the lag of Base's finalized head at 846–890 s on mainnet and 1,058–1,102 s on Sepolia.

3. Model and Goals

Parties. A taker T holds shielded ZEC and wants u tokens; a maker M holds the tokens. A probabilistic polynomial-time adversary $\mathcal{A}$ statically corrupts at most one of them. It schedules transactions within the bounds below and observes all public data. It may also crash the honest party at any time and restore it later; the party's state persists. We assume one thing about crashes:

- **(H) Continuity.** An honest party that has broadcast a transaction publishing or committing to its share stays online until that transaction's effect is final. For the taker this spans from her first `commitClaim` to the finality of her claim, or of the maker's refund.

Hash-locked swaps make the same assumption about the party that reveals a preimage. Without it a taker who commits and then goes offline past the reveal window cannot claim, and a malicious maker need never refund, so her ZEC stays locked. The commit signature has no validity interval, so a commit that reverted can be replayed later; under (H) this is harmless, because the taker is online and reveals within W. Rule L_M (Section 4) adds one requirement for the maker.

Clock. A party measures time by the timestamp of the latest EVM block it has seen, which we write τ. Block timestamps are monotone. The bounds below are stated in this clock, so a party's view lag and the sequencer's timestamp drift are part of D.

EVM ledger.

- **(E1) Bounded inclusion.** An honest transaction broadcast at τ is included in a block with timestamp at most $\tau + D$, unless a reorganization drops it. If its preconditions fail at inclusion, it reverts.
- **(E2) Finality and reorganizations.** At τ, blocks with timestamp at most $\tau - F$ are final. A younger block may be reorganized away; its transactions are dropped, and the sender must re-send them. $\mathcal{A}$ reorganizes at most R times per swap.
- **(E3) Public calldata.** $\mathcal{A}$ learns the calldata of every broadcast transaction, *including one that later reverts or is reorganized away*.

Each reorganization can come just before finality, after which the re-sent copy again needs up to D. A transaction that must survive R reorganizations therefore needs the budget

$$B_R = (R + 1)D + RF.$$

Zcash ledger.

- **(Z1) Persistence.** A transaction with k_Z confirmations is reverted with probability at most ε_Z. We do not quantify ε_Z; a deployment should choose k_Z from the value at risk, and in blocks rather than minutes, since NU7 shortens the block interval.
- **(Z2) Liveness.** A valid honest transaction is mined within Δ_Z, which includes any expiry (40 blocks after its target height) and re-send.

Cryptographic assumptions.

- Discrete logarithms are hard on Pallas.
- Halo 2 with the Ironwood action circuit (the circuit as fixed in NU6.2) is knowledge-sound and zero-knowledge.
- RedPallas spend authorization is unforgeable for keys rerandomized by additive offsets known to the adversary [2, 14]. Zcash relies on the same property, since every spender chooses its own randomizer α.
- The Fiat–Shamir Schnorr proofs of knowledge are straight-line extractable in the algebraic group and random oracle models [15].
- keccak256 and BLAKE2b behave as random oracles.
- ECDSA on secp256k1 is EUF-CMA.
- ChaCha20-Poly1305 is a secure AEAD (IND-CPA and INT-CTXT), and Gap-DH holds on Pallas.
- Ironwood transactions of the same *shape* are indistinguishable to an observer without the keys. The shape is every public field outside the cryptographic payload: the number of actions, the fee, `lock_time`, the expiry height relative to the mined height, and the block. Section 8.2 measures how large these classes are.
- Token transfers eventually succeed. A token issuer that blocks an address forever is an issuer risk, which no protocol over that token can remove.

Goals. Let v be the ZEC funded to the joint address and u the tokens escrowed.

Definition 1 (Taker safety)**.** An honest T ends in one of three states: (i) she never funded, and lost at most the

premium π (Section 5); (ii) she is credited $u - f_r$, where f_r is the relayer fee; or (iii) she can spend the joint note, or has spent it. In particular, no execution leaves her ZEC spent by someone else while she is uncredited.

Definition 2 (No hostage)**.** An honest T who funded reaches (ii) or (iii) within bounded time, without any action by M.

Definition 3 (Maker safety)**.** An honest M ends either refunded, or with T credited and M able to spend a final joint note of value at least v.

Definition 4 (Unlinkability)**.** A passive observer of both chains cannot distinguish a swap's Zcash transactions from other Ironwood transactions of the same shape in the same blocks.

4. The Protocol

4.1 Key setup

Quotes and requests travel over Zcash memos (Section 7); here we give the keys.

1. T samples $a \leftarrow \mathbb{F}_q^*$ and a seed s_T; M samples b and s_M, all fresh per request. They exchange $A = [a]\mathcal{G}$ and $B = [b]\mathcal{G}$, each with a Schnorr proof of knowledge. Each proof's challenge binds the request identifier sid and the party's role, and each party verifies the other's proof before use.
2. T commits to s_T in her request and opens it in her acceptance. Both derive $\mathsf{nk} = H_{\mathbb{F}_p}(\mathsf{dom}_{\mathsf{nk}}, \mathsf{sid}, s_T, s_M)$ and $\mathsf{rivk} = H_{\mathbb{F}_q}(\mathsf{dom}_{\mathsf{rivk}}, \mathsf{sid}, s_T, s_M)$ with personalized, length-prefixed BLAKE2b-512 and distinct domain tags.
3. If $\tilde{y}(A + B) = 1$, both parties negate their shares and points. This is deterministic, so $\mathsf{ak} = A + B$ then has $\tilde{y} = 0$. The escrow stores the normalized points; a proof of knowledge of $\log A$ is also one of $\log(-A)$.
4. Both check that $\mathsf{FVK}_J = (\mathsf{ak}, \mathsf{nk}, \mathsf{rivk})$ is a valid full viewing key, and derive the external address at index 0 as addr_J, encoded with an Orchard receiver only. Value sent there enters Ironwood.

Both parties can view addr_J; spending needs $a + b$. The proofs of knowledge prevent the rogue-key choice $B = [b']\mathcal{G} - A$ [16]. ZIP 2005 [6] specifies how wallets derive keys so that notes stay recoverable after a quantum break; both of its branches start from a spending key or a quantum spending key. A joint key has neither, so joint notes are not quantum-recoverable. Consensus does not check how a full viewing key was derived, and a joint note lives only until the swap settles.

4.2 Escrow

For each swap the escrow stores the maker, the taker's payout address, the token, u, f_r, A and B, deadlines $t_0 < t_1 < t_2$ and a reveal window W. The swap identifier is keccak($maker, salt$), so it is scoped to the maker. The salt is $H(\mathsf{dom}_{salt}, \mathsf{sid}, s_T, s_M)$. The request identifier sid travels in memo envelopes, so a salt derived from it alone would let anyone who later obtains a party's viewing key find the swap's escrow. The state machine is in Table 1.

Call	From → to	Condition and effect
`open`	– → Open	By M. A and B on the curve and canonical, $f_r \le u$, $\tau < t_0 < t_1 < t_2$. Pulls u tokens and rejects tokens that deliver less.
`setReady`	Open → Ready	By M, $\tau < t_0$.
`commitClaim`	Ready → Locked	ECDSA signature by the payout key over (escrow, chain id, swap id, c, relayer), $\tau < t_1$. Stores c = keccak(id, a); reveals nothing.
`commitClaim`	Open/Ready → Locked	Same signature, $\tau \ge t_2$ (punish path).
`claim`(a)	Locked → Claimed	Anyone. $\tau <$ *lockedAt* $+ W$, keccak(id, a) $= c$, $[a]\mathcal{G} = A$. Credits $u - f_r$ to the payout and f_r to the committed relayer. No external calls.
`refund`(b)	Open → Refunded	$\tau < t_0$, $[b]\mathcal{G} = B$. Credits u to M. No external calls.
`refund`(b)	Open/Ready → Refunded	$t_1 \le \tau < t_2$.
`refund`(b)	Locked → Refunded	$\tau \ge$ *lockedAt* $+ W$.
`withdraw`	–	Anyone; pays an owner's credit. Reveals nothing.

Table 1. Escrow state machine. Claimed and Refunded are terminal.

The Pallas check uses Jacobian coordinates and a projective comparison with the stored point. It rejects scalars equal to 0 or at least q, non-canonical coordinates, and the point at infinity. The escrow verifies a Pallas point directly, so unlike the Monero–Ethereum construction [29, 30] it needs no cross-curve equality proof.

4.3 Settlement and wallet rules

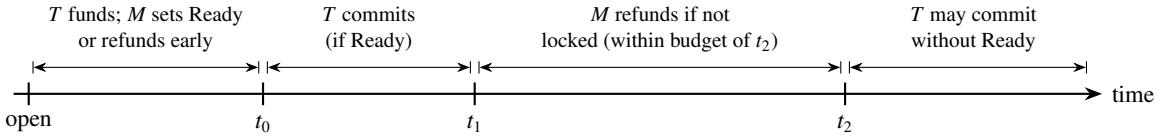


Figure 1. Deadlines. After a commit, T reveals within W; after W, M may refund a lock that was never revealed.

Happy path.

1. M opens the escrow.
2. Once open is final, T funds $v + f$ ZEC to addr_J in a fully shielded transaction.
3. After k_Z confirmations, M sets Ready.
4. T commits; once the lock is final, she reveals a by submitting `claim`(a), waits for the claim to be final, and withdraws.
5. M reads a, signs a spend of the joint note under $(a + b) + \alpha$ through PCZT, and sweeps it to himself with $\mathsf{ovk} = \perp$ and no change to addr_J.

In a refund, b is published and T sweeps the note back the same way.

Relayer. The payout address should hold no ETH, since funding it would link it to the funder. `commitClaim` therefore names a relayer, who submits the transactions and receives f_r. `claim` is permissionless and pays the committed relayer whoever submits it. A relayer that receives a could withhold it and publish it after the window closes; T would then lose both legs. Rule R2r prevents this with a fallback: if the relayer has not broadcast the claim within a hand-off time D_h (part of D), T submits it herself, from any funded address, or through a second relayer. The budget B_R is counted from the hand-off. With the fallback, a relayer can cost T gas and some privacy but not funds. Without it, the relayer is trusted for safety. Our implementation self-relays: the taker holds the relayer key.

Table 2 lists the wallet rules. They are part of the protocol: the proofs in Section 5 depend on them. Section 6 shows an attack when any of R2c, R2r, R3e, R3d, R3l or L_M is removed. R1 and R1b protect the taker's funding against a misconfigured or late escrow, R1q against windows the maker chooses, and R4 protects against a later compromise of the device; the model does not exercise these three.

Rule	Party	Statement
R1	T	Fund only after open is final and every stored parameter equals the quote.
R1q	T	Accept a quote only if $W > F + B_R$, computed with T's own D, F and R, and $t_2 - open \le H$ for a horizon H that T chooses. The escrow enforces neither.
R1b	T	Fund only if $\tau + \Delta_{fund} < t_0$, where Δ_{fund} covers k_Z confirmations at a high percentile of block intervals (block times are random), plus D for `setReady`.
R2c	T	Commit only once Ready is final and $\tau + B_R < t_1$; or from t_2 if the state is Open or Ready. (A late commit reverts and reveals only c.)
R2r	T	Reveal a only once the lock is final and $\tau + B_R < lockedAt + W$, with τ the hand-off time. Re-send the claim whenever it is dropped; fall back to self-submission if the relayer has not broadcast it.
R3e	M	Refund early only in Open with no sufficient note, after T's R1b funding window has closed. Once b is public, never set Ready.
R3d	M	Refund after t_1 only if a final block with timestamp $\ge t_1$ shows Open or Ready, and $\tau + B_R < t_2$. Re-send whenever dropped.
R3l	M	Refund a lock only if a final block with timestamp $\ge lockedAt + W$ shows Locked. Re-send whenever dropped.
R4	both	Per-request secrets are fresh. Each is erased only once nothing can still need it: after the party's refund or claim is final and its sweep has k_Z confirmations (an expired sweep is re-sent). A taker who broadcast a funding keeps a until that funding is swept, or has expired unmined by k_Z blocks, even if the maker has refunded and her wallet does not yet show the note.
L_M	M	**Maker liveness:** online at some $\tau \in [t_1 + D + F,\ t_2 - B_R)$, and then, by (H), until its refund is final.

Table 2. Wallet rules. $B_R = (R + 1)D + RF$ is the reorganization budget of Section 3.

Parameters. The rules need

$$W > F + B_R, \qquad t_0 - open > F + \Delta_{fund}, \qquad t_1 - t_0 > F + B_R, \qquad t_2 - t_1 > D + F + B_R.$$

The bounds on W and $t_2 - t_1$ carry the safety proofs (Section 5); those on t_0 and t_1 only make the happy path possible. TLC is consistent with the bounds at three parameter points, all with $D = F$, and finds an attack when either budget is one step short (Section 6). Our implementation writes the R2r and R3d tests as $\tau + D + F + \mu < d$ and refuses to start unless $\mu \ge RD + (R - 1)F$, which is the same condition. The maker writes the quote, so the taker checks the bound on W herself (R1q), with her own D, F and R; the maker refuses a $t_2 - t_1$ below its bound. Without R1q a maker could quote $W \le F + B_R$, so that a final lock can never be revealed safely and only his refund, which publishes b, could end the swap; or he could place t_2 so far ahead that the punish path holds the taker's ZEC indefinitely. Our taker uses $H = 48$ h. The TLA$^+$ model fixes W at each point; R1q is what keeps every accepted quote within a checked bound. Table 3 gives a profile for Base mainnet.

Parameter	Base mainnet, live sequencer
D (inclusion bound)	10 min
F (finality lag; measured about 15 min)	20 min
R (reorganizations survived)	2
$B_R = (R + 1)D + RF$	70 min
k_Z	10 blocks (12.5 min at 75 s; 4.2 min under NU7)
$t_0 - open$	1 h (+ $3X$ with cover delays)
$t_1 - t_0$	1.75 h
W	1.75 h
$t_2 - t_1$	2 h + maker downtime tolerance (e.g. 24 h)

Table 3. Example parameters. They assume a live sequencer that includes honest transactions within D. During a sequencer outage neither D nor F is bounded on this time scale, and block timestamps can lag real time; we do not cover that case (Section 11).

5. Security Analysis

Lemma 1 (Key isolation)**.** *Suppose a corrupted party's share point X_c is accepted with a valid proof of knowledge bound to* (sid, *role*)*, and the honest share x_h is never published. Then $\mathcal{A}$ produces a transaction that spends a note*

at addr_J *with negligible probability.*

Proof sketch. We reduce to the unforgeability of RedPallas under additively rerandomized keys. The reduction $\mathcal{B}$ receives the challenge key X_h and embeds it as the honest share. It simulates the honest party's proof of knowledge by programming the random oracle, and extracts x_c straight-line from $\mathcal{A}$'s proof (algebraic group model), so $\mathsf{ak} = X_h + [x_c]\mathcal{G}$. Spend-authorization signatures needed by the honest party are obtained from the signing oracle under the corresponding offsets. Suppose $\mathcal{A}$ outputs a valid spend. Knowledge soundness of Halo 2, applied once to that final transaction, gives α with $\mathsf{rk} = \mathsf{ak} + [\alpha]\mathcal{G}$. The spend's signature is then a forgery for X_h under the known offset $x_c + \alpha$ on a fresh sighash. No rewinding is nested inside another extraction. Sign normalization negates (x_h, x_c) together and preserves the argument. Without the proof of knowledge, $X_c = [x']\mathcal{G} - X_h$ would give $\mathcal{A}$ the whole key; this is the rogue-key attack the proof prevents. □

Lemma 2 (Budgeted transactions)**.** *Let an honest party broadcast a transaction at τ with $\tau + B_R < d$, re-send it whenever it is dropped, and stay online until a copy is final. Then a copy is included in a block with timestamp before d and is never removed.*

Proof. By E1 the first copy is included by $\tau + D$. A copy included at x can be reorganized only before $x + F$ (E2), and the re-sent copy is then included by $x + F + D$. So the k-th copy is included by $\tau + D + (k-1)(D+F)$. After at most R reorganizations the last copy is included by $\tau + D + R(D+F) = \tau + B_R < d$ and can no longer be removed. A copy that is not reorganized stays. □

Lemma 3 (Honest revelations succeed)**.** *(a) A claim sent under R2r becomes final as* Claimed*. (b) A refund sent under R3d or R3l becomes final as* Refunded*. (c) Under R3e, b becomes public only in* Open *before Ready, and the swap never becomes* Ready *afterwards.*

Proof sketch. By Lemma 2 it suffices that each precondition holds at every inclusion before the deadline.

(a) The lock is final, so the state stays Locked until a claim lands or the window closes; the only other exit from Locked needs $\tau \geq \mathit{lockedAt} + W$. The claim makes no external calls, so token behavior cannot make it revert. With the relayer fallback of R2r, the first copy is broadcast within the budget even if the relayer withholds it.

(b) For R3d: the final block with timestamp $\geq t_1$ shows no lock. Timestamps are monotone, so every block before t_1 is final and contains no successful commit. In $[t_1, t_2)$ commits are disabled, and only M can set Ready, which it does not. So Open or Ready with $t_1 \leq ts < t_2$ holds at every inclusion before t_2. A refund that is merely included before t_2 could be reorganized past it; Section 6 shows the resulting attack. For R3l, a final lock past its window can no longer be claimed.

(c) Immediate from R3e. □

Theorem 1 (Taker safety and no hostage)**.** *Let T be honest and satisfy (H) and R1q, so that she is online from her commit until it is resolved and every window she accepted meets its bound, with no assumption on M, and let the relayer be honest or the fallback of R2r be available. Except with probability* $\mathsf{negl} + \varepsilon_Z$*, the protocol satisfies Definitions 1 and 2. If M never refunds, T's claim is final by $\tau_T + B_R + W + F$, where $\tau_T \geq t_2$ is the first time from t_2 on that T is online; for a taker online at t_2 this is $t_2 + B_R + W + F \leq open + H + B_R + W + F$. (H) does not oblige T to be online before her commit, so a taker who is offline at t_2 is resolved only after she returns; her ZEC is not lost meanwhile, because refunds from* Open *and* Ready *are disabled from t_2. If M refunds, T's sweep back has k_Z confirmations within $C + \Delta_Z + k_Z\beta$ of the refund, where β bounds a block interval at a chosen percentile and C is the cover delay, capped as in Section 8.2.*

Proof sketch. By Lemma 1, only someone who knows $a + b$ can spend the joint note. M learns a only from T's claim, which by Lemma 3(a) credits her (ii). If M refunds, b is public, and T sweeps (iii); by R4 she keeps a until the sweep is confirmed, and re-sends it if it expires. If neither happens by t_2, rule R2c lets T commit once she is online, at τ_T: refunds from Open and Ready are disabled from t_2, so M cannot block the commit, which lands by $\tau_T + B_R$. She then reveals within W, which R1q guarantees exceeds $F + B_R$, and Lemma 3(a) gives (ii). R1q also bounds t_2. Each step uses only T's own actions. If M never opens, T does not fund (R1) and has lost at most π. □

Theorem 2 (Maker safety)**.** *Let M be honest and satisfy L_M and (H), with no assumption on T. Except with probability* negl $+ \varepsilon_Z$*, the protocol satisfies Definition 3.*

Proof sketch. T is credited only through Locked, which is reached in one of two ways:

- Ready before t_1. M sets Ready only after a note of value at least $v + f$ has k_Z confirmations (Z1), and never after publishing b (Lemma 3(c)).
- $ts \geq t_2$ from Open or Ready. By L_M, M sends an R3d refund at some $\tau < t_2 - B_R$, and by Lemma 3(b) it is included before t_2 and never removed, so this case cannot occur.

If T is credited, M knows $a + b$. T does not know b: an R3d or R3l refund succeeds and ends the swap, and an R3e refund publishes b only in a swap that never becomes Ready. By Lemma 1 no one else spends the note, and M sweeps it, keeping b until the sweep is confirmed (R4). □

Proposition 1 (L_M is needed)**.** *If M is offline throughout $[t_1, t_2)$, a taker who never funds can wait until t_2, commit from Open, and claim u. The same holds for any protocol in which the EVM side cannot observe the Zcash funding and the taker's unilateral exit after a deadline is paid on the EVM side: an absent maker cannot be protected after that deadline. A design whose exit is on the Zcash side, such as a timed commitment to b [31], avoids L_M at the cost of time-lock puzzles; we do not claim otherwise. An escrow that verified a proof of funding against a Zcash light client would not need L_M, at a large cost in complexity.*

TLC exhibits the attack (configuration c′ in Section 6), and harness scenario s11 executes it (Section 9). Gugger's punish window [27] has the same requirement.

Theorem 3 (Liveness)**.** *If both parties are honest and online and the parameter conditions hold, the swap ends with T's claim or M's refund. It ends with the claim, final by $t_1 + W + F$, and M's sweep has k_Z confirmations within $C + \Delta_Z + k_Z\beta$ after it, if* `setReady` *is included before t_0 and not removed by a reorganization. R1b budgets only D for* `setReady` *and a percentile for the funding's confirmations, so this is likely but not guaranteed: a reorganization that removes* `setReady` *just before t_0 leaves its re-sent copy too late, and the maker refunds.*

Free options. The taker's price is fixed when she accepts, and until she funds she can walk away at the cost of the premium; after Ready she may choose not to commit, and after committing not to reveal. Her option therefore runs from the Accept to her last reveal, up to $\theta = (t_1 - t_{\text{Accept}}) + W$. Before Ready, M may refund. Han, Lin and Yu model atomic swaps as premium-free American call options [32]. Cloakroom charges a non-refundable ZEC premium π with the taker's acceptance. An at-the-money option with volatility σ_u of the ZEC/token exchange rate over time θ is worth about $0.4\,\sigma_u\sqrt{\theta}\,u$. For USDC, σ_u is ZEC's dollar volatility. At $\sigma_u = 100\%$ per year and the profile of Table 3 without cover delays, $\theta \approx 1 + 1.75 + 1.75 = 4.5$ h and the option is worth about 0.9% of u; with the private cover profile of Section 8.2, the maker's delay before `open` adds about 2.6 h and $t_0 - open$ must absorb three draws of the taker's funding delay (about 8.75 h), so $\theta \approx 15$ h and the option is worth about 1.7%. The premium should cover this plus the maker's `open` and `refund` gas. Because the premium is paid before `open`, a maker could take premiums and never open; the taker then loses π, the bound in Definition 1(i), and holds the maker's signed quote as evidence. The maker's own option before Ready is not priced; it ends at t_0, and an early refund is public.

6. Mechanized Verification

6.1 Protocol model in TLA$^+$

The model covers:

- the escrow guards of Table 1;
- an EVM block history with finality lag F, reorganizations of non-final blocks that drop their transactions, and a reorg budget R;
- inclusion within D, with reversion when a precondition fails at inclusion;
- secrets that leak at broadcast, even when the transaction reverts or is reorganized away;
- a Zcash note that may be reorganized before k_Z confirmations;

- an adversary that corrupts at most one party and controls scheduling and crashes;
- honest parties that act on finalized state and re-send dropped transactions.

We check four invariants:

- NoDoubleExit;
- NoFreeClaim (a claim implies a final funding existed);
- HonestTakerSafe;
- HonestMakerSafe.

We also check two liveness properties under weak fairness: SwapSettles and NoHostage.

Time is discrete, and a block can be reorganized only while its age is below F steps, so in the model the budget of Section 3 is $D + R(D + F - 1)$. At $D = 1$ this equals $D + RF$, the margin our first version used, which omits one D per reorganization. Our first checks ran only at $D = F = 1$, where the two coincide, and missed the error; a reviewer found it. We now check a second point, $D = F = 2$, where they differ.

Point	Configuration	Result	Distinct states
$D = F = 1$	(a) both honest	all invariants hold	352
	(b) malicious maker	all hold	52,423
	(c) malicious taker, maker satisfies L_M	all hold	327,390
	(c′) as (c), maker offline in $[t_1, t_2)$	NoFreeClaim fails	–
$D = F = 2$	(a) both honest	all hold	2,406
	(b) malicious maker	all hold	856,632
	(c) malicious taker, maker satisfies L_M	all hold	13,467,219
	(c″) as (c), maker online only from L_M's last moment	all hold	2,124,904
	liveness, both honest	SwapSettles holds	2,406
	liveness, malicious maker	NoHostage holds (punish path)	856,632
$D = F = 2$,	(a) both honest	all hold	9,626
$R = 2$	(b) malicious maker	all hold	3,588,748
	(c) malicious taker, maker satisfies L_M	all hold	120,580,544
	(c″) maker online only from L_M's last moment	all hold	9,075,177

Table 4. TLC 1.7.4 results, with $k_Z = 1$ and $R = 1$ unless stated. At $D = F = 1$: $t_0 = 4$, $t_1 = 7$, $t_2 = 13$, $W = 4$. At $D = F = 2$: $t_0 = 8$, $t_1 = 15$, $t_2 = 25$, $W = 8$, claim budget 5 and refund margin 2 (the maker's first refund goes out by $t_2 - D - F - 2$). These are the bounds of Section 4 in the model's discrete form, where time moves in whole steps and the strict bounds become $\geq$. At $R = 2$: $t_0 = 11$, $t_1 = 21$, $t_2 = 34$, $W = 11$, claim budget 8 and refund margin 5. Liveness assumes an online taker, which is assumption (H).

Mutations. We removed one rule at a time, or shortened one budget by one step, and TLC finds a counterexample for each:

- *Claim budget one step short* ($D = F = 2$, at $R = 1$ and at $R = 2$): the claim is reorganized and its re-sent copy lands after the window, so it reverts, publishing a, and the taker loses both legs.
- *Refund budget one step short* ($D = F = 2$, $R = 2$, maker online only from the last moment the shortened L_M allows): the refund is reorganized, its re-sent copy lands after t_2, and a taker who never funded claims through the punish path. A maker who is online throughout refunds at $t_1 + D + F$ and is not affected; the budget protects the maker who comes back late.
- *No re-send of a dropped refund after its first deadline* ($D = F = 2$): the same free claim. Our implementation had this bug; it is fixed.
- *Ready after the maker has published b* ($D = F = 2$): an early refund near t_0 and a late funding by a malicious taker overlap. Ready lands, the taker locks and claims, and she also sweeps the ZEC with b. Our

implementation already obeyed this rule, but the model and the paper did not state it.

- *Reveal before the lock is final:* a reorganization drops the lock, the claim reverts and publishes a, and the taker loses both legs.
- *Claim from* Open *without Ready* (our first design): free claim.
- *No punish window:* permanent hostage (liveness failure; the ZEC is locked, not stolen).
- t_2 *too close to* t_1, with all rules intact and an online maker (described below).
- *Refund without finality,* in two variants: a free claim through the punish path.

The t_2 mutation was our first finding. With $t_2 - t_1 = 4$ at $D = F = 1$:

1. The maker refunds at the earliest moment his rules allow, $t_1 + D + F$.
2. The refund is included but not final at t_2.
3. The adversary reorganizes it away, and the escrow is Open again.
4. A taker who never funded commits through the punish path and claims.

Rule R3d therefore requires the refund to stay within the budget before t_2, and $t_2 - t_1$ must leave room for that. The model abstracts cryptography to the information each broadcast reveals, and it checks three small parameter points, $(D, F, R) \in \{(1, 1, 1), (2, 2, 1), (2, 2, 2)\}$ with $k_Z = 1$; the inequalities of Section 4 are argued in general. Because every point has $D = F$, the model cannot tell $(R + 1)D + RF$ from $RD + (R + 1)F$; that the budget has the first form rests on Lemma 2. The liveness checks assume an online taker; a taker who commits and goes offline past W is outside them, as (H) states.

6.2 The escrow contract

We verified the final contract in four ways; logs and commands are in the repository's audit report.

- **Symbolic execution (Halmos 0.3.3).** Proved with symbolic arguments, callers, times and pre-states:
  - each of `open`, `setReady`, `commitClaim`, `claim` and `refund` succeeds if and only if its guard holds, with exactly the specified effects and credits;
  - every call moves a swap only along the state machine's edges, and terminal states are absorbing;
  - no claim door and refund door are ever open at the same instant;
  - after t_2, an Open or Ready swap can be locked but not refunded;
  - swap identifiers are scoped to the maker;
  - `claim` and `refund` make no external calls;
  - state is written before every token call, even when a malicious token re-enters with arbitrary calldata.

  Conservation, bounded credits and supply hold after every sequence of up to two calls on any swaps, up to four calls on one swap, and along a directed six-call claim path. Unrestricted three-call sequences did not terminate and are not claimed. Four reachability witnesses show the specifications are not vacuous. Here the curve check is an uninterpreted predicate, so the proofs hold for any curve arithmetic.
- **Curve arithmetic.** The exact source of the Pallas library, with only its constants replaced, was checked exhaustively on two small prime-order curves with $a = 0$ and $Q > P$ (229 and 5,023 points). For every scalar k and point (x, y), `isDiscreteLogOf`(k, x, y) holds iff $0 < k < Q$ and $[k]G = (x, y)$. Small curves cannot exercise 256-bit word effects, so we also argue those directly. The library uses only `mulmod` and `addmod`, which are exact for all 256-bit operands, and subtractions $p - x$ with $x < p$; there is no lazy reduction. On Pallas itself, test vectors are shared with the Rust implementation and the group law is fuzzed. A structured test covers 514 boundary scalars (2^i, $q - 2^i$, $2^{254} \pm 1$ and scalars just below q), each checked against independent affine arithmetic: $[k]G + [q - k]G$ is the identity, doubling agrees, and the alias $k + q$ is rejected.
- **Mutation testing.** All 22 seeded bugs were caught: guards dropped or loosened, the punish window moved,

effects placed after interactions, each audit fix reverted. Two curve-guard mutants are beyond symbolic execution and were caught by concrete tests. One of them revealed a missing test for alias scalars $k + Q$, which we added.

- **Stateful fuzzing and static analysis.** Five invariants (conservation, bounded credits, exclusive and absorbing exits, a per-transition oracle, supply) held over 10^4 runs of depth 100 each, against the unstubbed contract. Slither 0.11.6 (102 detectors) reported 16 results. One was ranked high: the balance read before the token pull in `open`. It is not exploitable, because every re-entrant escrow operation is balance-exact on its own. The rest concern timestamps, log order and naming.
- **Findings, both fixed.** L-01 (low): `open` credited the stated amount, so a fee-on-transfer token could leave the escrow under-collateralized. `open` now requires the balance to rise by exactly the amount. I-01 (informational): the Pallas check relied on its caller for coordinate canonicality. It now checks itself.

Transaction	**Gas used**
`open` (two on-curve checks, token pull)	337,254
`setReady`	32,494
`commitClaim` (signature recovery)	69,456
`claim` (scalar check $[a]G = A$)	555,423
`refund` (scalar check $[b]G = B$)	529,429
`withdraw`	52,259–57,071

Table 5. Gas per transaction from receipts, including intrinsic and calldata cost. The scalar multiplication dominates `claim` and `refund`.

7. Messages

A maker publishes a unified address, an EVM address and an RFQ public key for the current epoch, with an EIP-191 signature by the EVM key over the other three. A taker refuses an entry whose signature does not verify, so no one can advertise a known maker's EVM address with his own Zcash address and collect premiums. Each message is a 512-byte memo on a zero-value shielded output (ZIP 302 type `0xFF` [11]) and is encrypted end to end on top of Zcash's note encryption:

- the RFQ is encrypted to the epoch key;
- the Quote and the Accept are encrypted under a Diffie–Hellman exchange of per-request ephemeral keys, using ChaCha20-Poly1305 with keys derived from BLAKE2b.

Message	**Encrypted payload**	**Memo bytes**
RFQ	id, lots, A, π_A, commitment to s_T, payout address, reply address	269
Quote	price, relayer fee, premium, B, π_B, s_M, escrow, maker address, $t_0/t_1/t_2/W$ offsets, maker's EVM signature	346
Accept	s_T, hash of the quote (sent with the premium as value)	105

Table 6. Messages; only the type, an identifier and ephemeral keys are visible to the recipient's viewing key.

The maker signs each quote with his EVM key, over the request identifier and the quote's contents. The taker verifies the signature before paying the premium and keeps it, so a maker who takes a premium and never opens leaves transferable evidence. The taker also checks on chain that the escrow was opened by the maker's EVM address with the quoted parameters.

8. Privacy

8.1 What each observer learns

Theorem 4 (Unlinkability)**.** *Under the assumptions of Section 3, there is a simulator that, given only the swap's public EVM data (A, B, c, one share, the payout and relayer addresses, u, f_r and the times) and the shapes (Section 3) and block heights of its Zcash transactions, outputs Zcash transactions indistinguishable from the real ones.*

Proof sketch. The simulator outputs random Ironwood transactions of the same shapes. Each real spend publishes $\mathsf{rk} = \mathsf{ak} + [\alpha]\mathcal{G}$ for a fresh uniform α, which is uniform and independent of ak. Note commitments are hiding. Nullifiers are pseudorandom under nk, which is never published. Note ciphertexts reveal nothing without the keys, and Halo 2 proofs are zero-knowledge. A hybrid argument over these components gives the claim. □

The theorem is only as strong as its shape assumption. Fee, expiry and `lock_time` are public and do fingerprint wallets, so the shape includes them, and Section 8.2 measures the class our wallet falls into. The observer is then reduced to correlating times.

Composition. One swap makes five Zcash transactions: the RFQ, the Quote, the Accept with the premium, the funding, and one sweep (by the maker after a claim, or by the taker after a refund). Public EVM events anchor them from both sides. The funding follows `open`, the sweep follows `claim` or `refund`, the Accept precedes `open`, and the funding precedes `setReady`. An observer can look forward from an event or backward from it. Every party therefore applies the cover delay of Section 8.2 on both sides of each public event: the maker before `open` and before `setReady`, the taker before funding, and both before their sweeps. Without the delay before `setReady`, the funding would sit exactly k_Z blocks before it. The delays are independent, but the observer's posteriors are not: the funding lies after `open` and the taker's delay and before `setReady`, k_Z blocks and the maker's delay, and the two windows together can narrow it well below either alone. Table 8 measures only one trigger at a time; we do not measure the joint posterior. Linking the swap needs only one of its transactions, so the per-swap probability is higher than the per-transaction figures. All of a maker's swaps are linked through his public EVM address. An intersection attack across them would still need a feature that persists across his shielded transactions; none is exposed beyond the wallet fingerprint, which a large group shares.

Counterparties and infrastructure.

- *The maker* learns the requested size, the funding transaction and the taker's fresh addresses, but not her identity or balances.
- *The taker* can see that the joint note was spent, since she knows nk, but not where it went: the sweep uses $\mathsf{ovk} = \bot$. We verified that with the shared ovk instead, the counterparty recovers the destination.
- *The relayer*, if it is not the taker herself, sees a shortly before it is published, together with the network address she sends it from. Without Tor this links her IP address to the swap.
- *A light-wallet server* sees the client's address and which full transactions it fetches. Compact blocks carry no memos, so our wallet fetches each transaction that pays it. This leak should be closed with Tor and decoy fetches, which we have not implemented.

Later key compromise. All outputs of swap transactions, including messages, funding and change, are created with $\mathsf{ovk} = \bot$, so a party's outgoing viewing key reveals none of them. Consider an adversary who later obtains a party's incoming viewing key, after the R4 erasure and the epoch-key rotation. It learns which incoming notes carried Cloakroom messages, their times, their types, the request identifier and the ephemeral keys in the clear part of each envelope, and the value of each received note, which for the maker includes the premium. It does not learn the encrypted payloads: shares, seeds, prices, addresses and deadlines (Gap-DH and AEAD security). The request identifier does not lead to the escrow, because the escrow salt depends on both seeds (Section 4.2). Before rotation, the epoch key exposes RFQ contents. Our implementation rotates the epoch key manually.

Amounts. ZEC amounts never appear on chain. The USDC amount is public by design, and lots of fixed size make the ZEC amount inferable from it. The maker's EVM address links all of his trades, which is inherent to a public market maker.

8.2 Measured anonymity sets

We scanned every Zcash mainnet block from Ironwood's activation to October 14, 2026 through the public `zec.rocks` lightwalletd server. The primary window is the last 30 days: heights 3,483,191–3,517,592 (34,402 blocks, mean interval 75.3 s). Cloakroom's transactions are fully shielded, Ironwood-only and have exactly two actions. This *structural fingerprint* covers 56,361 transactions: 33.1% of Ironwood transactions, and 68.3% of Ironwood transactions have two actions. Over the same period, 51.0% of Ironwood transactions touched the transparent pool (25.4% deshielded). Compact blocks omit fees and expiry heights. In a sample of 1,500 full transactions of this shape, 92.8% paid exactly our 10,000-zatoshi fee, and 55.7% matched our fee, `lock_time` and expiry together. Expiry height is a wallet fingerprint, and our default of 40 blocks is the largest group.

Window after a trigger	p10	median	p90	median, full settings
same block	0	0	4	0
5 min	0	3	14	1.9
15 min	3	8	39	5.5
60 min	16	33	158	22.3
24 h	492	847	4,223	535

Table 7. Number of transactions with the structural fingerprint in a window (Zcash mainnet, 30 days). The last column also requires our fee, `lock_time` 0 and an expiry 30–40 blocks after the mined height. It weights each transaction by the match rate measured for its block-load class (76.8% in blocks with one look-alike, down to 45.4% in blocks with 16 or more; 55.7% overall). Quiet periods match more often, so the ratio to the structural column exceeds 55.7%.

Without delay, a swap transaction that follows an observable trigger, such as an `open` or a `claim` on Base, hides among almost no others. Traffic is bursty, averaging 78 matching transactions per hour but with a median hour of 33, and the first month after activation saw only 7.3 per hour. Against a delay drawn uniformly from $[0, X]$, an anonymity set of $k = 50$ for 90% of transactions needs $X \approx 175$ min for the structural fingerprint and 240 min for full settings. For $k = 200$ it needs 615 and 875 min.

Cover policy. Before each broadcast on either side of a public event (above), the party waits for a random delay; the RFQ needs none. How the delay is drawn matters more than its length. Our first policy waited for a random delay *and* for k look-alikes since the trigger, broadcasting as soon as both held. An observer who knows the policy then knows that, whenever the count binds, the swap transaction comes right after the k-th look-alike. We estimated what a policy-aware observer learns. Let $f(b)$ be the probability that the policy's broadcast lands in block b, given the observed arrivals; we estimate it with 2,000 draws of the policy's randomness per trigger. The observer sees the look-alikes in each block, ours included. With a locally smooth background rate, the posterior that a given observed look-alike in block b is ours is proportional to $f(b)$. For 1,000 random triggers in the 30-day window we report the posterior mass on the true transaction (the success probability of an observer who guesses by sampling from its posterior; an observer who picks the most likely candidate does at least as well, and we do not report that figure) and the effective anonymity set 2^H, where H is the posterior's entropy. Candidates are structural look-alikes; full-settings matching would shrink the sets by roughly the factor in Table 7.

Parameters	Policy	Posterior on ours: median, p90	2^H: p10, median	Mean delay
$X = 175$ min, $k = 50$	uniform	0.011, 0.020	43, 79	88 min
	snap (first design)	0.177, 0.869	1.7, 21	121 min
	redraw (ours)	0.014, 0.026	35, 66	156 min
$X = 60$ min, $k = 20$	uniform	0.029, 0.060	14, 28	30 min
	snap	0.264, 1.000	1.0, 7.2	47 min
	redraw	0.036, 0.080	12, 26	60 min

Table 8. What a policy-aware observer learns about one delayed broadcast (Zcash mainnet, 30 days; 1,000 triggers). Uniform: delay $U \sim [0, X]$. Snap: $\max(U,$ arrival of the k-th look-alike). Redraw: wait U; if fewer than k look-alikes have appeared since the trigger, draw a fresh U and wait again.

Snapping concentrates the posterior: in one trigger in ten the observer identifies the transaction with near certainty. A uniform delay gives the most privacy for a given delay, but it has no floor when traffic is thin, as in Ironwood's first month (7.3 look-alikes per hour). Our wallet therefore uses *redraw*: the broadcast always ends a random wait, and at least k look-alikes have appeared since the trigger. It costs privacy close to uniform's for a longer delay. Its delay has an unbounded tail, which public deadlines cut. A wait cut at a deadline leaks that the broadcast came at the deadline, so the taker rejects quotes whose $t_0 - \mathit{open}$ cannot absorb three draws on top of F and D, and the maker's wait before `setReady` stops early enough for Ready to land before t_0. An attacker can inflate the count with its own look-alikes (about 0.02 ZEC for $k = 200$); this only weakens the floor, since the delay remains random. The wallet builds each transaction at broadcast time, so its expiry does not reveal the delay. Each Zcash transaction of a swap has two actions; the wallet refuses to spend more than two notes in one, so the funding never leaves the class. The private profile, $X = 175$ min and $k = 50$, adds about 2.6 hours on average per delayed step. Before `open` the maker's delay is part of the quote's lifetime, and the taker sets her quote timeout accordingly. Over a swap the delays add several hours, which is the price of the anonymity above and raises the option value the premium must cover.

9. Implementation and Evaluation

Implementation. A Rust node (`orchard` 0.16.0, `pczt` 0.10.0-pre.1, `zcash_primitives` 0.31.0-pre.1) implements key setup, messages, maker and taker state machines, and a light wallet that builds v6 transactions through PCZT and signs joint spends externally. It syncs from a Zebra node or from a lightwalletd server starting at the server's Ironwood tree state, and drives the EVM through Foundry's `cast`. Parties persist their state after every step and can be stopped and restarted at any stage. In one respect it is looser than the protocol: the taker commits as soon as Ready is visible, not once it is final (R2c). This is safe because a commit publishes only c, a commit on a Ready that is later removed reverts, and a is revealed only once the lock is final (R2r), which implies a final Ready. The model checks the stricter rule.

Against the production verifier. Against `orchard` 0.16.0 with the current (post-NU6.2) circuit:

- the joint full viewing key is accepted;
- a spend signed with one share is rejected;
- a spend signed with $(a + b) + \alpha$ passes proof, spend-authorization and binding-signature verification;
- with $\mathsf{ovk} = \perp$ the counterparty cannot recover the sweep's destination;
- a rogue share fails its proof of knowledge.

On a Zebra node. On Zebra 7.0.0-rc.0 in regtest, a v6 transaction with the real ZIP 244 sighash and an external signature under $(a + b) + \alpha$ was mined. A copy differing only in its spend signature, made with b alone, was rejected.

Two-party scenarios. Each scenario runs maker and taker as separate processes, which communicate only through encrypted memos on Zebra regtest and the escrow on a local EVM chain. Time is advanced on the EVM chain where needed, and in two scenarios the harness reorganizes the EVM chain to drop a party's transaction before it is final. Balances are checked exactly to the zatoshi, and every Zcash transaction is read back from the node. On the final code all eighteen scenarios pass, with 121 of 121 checks (about 48 minutes in total). Every scenario in which a party transacts (all but s15) reads back every Zcash transaction either party made and checks that it is a fully shielded v6 Ironwood transaction with no transparent inputs or outputs. Each of those scenarios also checks that each party that reached a terminal state has erased its per-swap secrets. A party that disappeared mid-swap keeps them, as it must in order to resume.

	Scenario	Observed outcome (checks)
s1	Happy path, with encrypted messages and premium	settled (10/10)
s2	Taker accepts, then disappears before funding	maker refunds before t_0, keeps the premium (7/7)
s3	Taker funds, disappears, returns after t_1	R2c declines to commit; maker refunds; taker recovers (8/8)
s4	Maker disappears after `open`	taker completes via the punish path; maker later sweeps (7/7)
s5	Maker sends a rogue key	quote rejected; nothing funded (4/4)
s6	Taker underfunds	no Ready; refund; taker recovers (7/7)
s7	Happy path with NU7 active	settled (10/10)
s8	Unfunded claim while the maker is offline; maker returns in $[t_1, t_2)$	lock rejected; maker refunds (6/6)
s9	Commit after t_1	reverts revealing only c; refund; taker recovers (7/7)
s10	Lock without reveal	maker refunds after W; taker recovers (7/7)
s11	Maker offline for all of $[t_1, t_2)$; taker never funds	free claim, as Proposition 1 predicts (6/6)
s12	Cover traffic with $k = 4$ (redraw policy)	each delayed broadcast ends a random wait with $\geq k$ look-alikes since its trigger (6/6)
s13	Reorganization drops the block with the taker's claim	taker re-sends; claim final; maker sweeps; one Claimed event (9/9)
s14	Reorganization drops the block with the maker's deadline refund	maker re-sends; finishes when final; taker recovers (9/9)
s15	Directory entry names the maker's EVM address but is signed by another key	taker refuses before any RFQ or premium; in s1 the genuine entry is accepted (2/2)
s16	Maker quotes a reveal window $W \leq F + B_R$	R1q rejects the quote; nothing funded, no premium (4/4)
s17	Maker quotes a punish deadline t_2 about 136 years ahead	R1q rejects the quote; nothing funded, no premium (4/4)
s18	Funding held unmined until after the maker's no-funding refund	taker keeps a (R4), then sweeps the late note with $a + b$ (8/8)

Table 9. Scenario suite on the final code; every check passed.

Public networks. On Zcash testnet (NU7 rules) and Base Sepolia, the first version of the protocol completed one swap and one refund between independent processes. All eleven Zcash transactions (one transfer that funded the taker's wallet and five per swap) were fully shielded. These runs exposed four problems that local chains did not, all fixed: a stale nonce, a limit on log queries, crashes on transient errors, and stale balance reads. The final escrow is deployed on Base Sepolia at `0xCCE4aFc81A6ccC9Db05B2a7D157942435b944283`, with runtime bytecode identical to our build of the internally audited source. Earlier versions of the wallet completed two swaps on it, with claims at Sepolia blocks 48,120,719 and 48,282,560. The final code then ran two swaps with $D = 30$ s, $F = 1{,}500$ s, $R = 1$, $t_1 - t_0 = t_2 - t_1 = W = 55$ min and $k_Z = 3$, which meet the bounds of Section 4 for $R = 1$ only; cover delays were off. In the first, with $t_0 = open + 40$ min, testnet mined the taker's funding 17 blocks after its broadcast. The maker, seeing no note at t_0, correctly refunded and published b, but the taker, whose wallet did not show the note either, concluded that nothing had been funded and erased a. The note was then mined, and its 0.0101 TAZ is unspendable. Rule R4 now keeps a until a broadcast funding is swept or has expired unmined, and scenario s18 reproduces the case; local chains had never delayed a funding. The second swap, with $t_0 = open + 80$ min, settled: RFQ, signed Quote and Accept with a 20,000-zatoshi premium, `open`, funding, Ready, commit, reveal and a claim at block 48,290,629, whose payout was withdrawn once it was final, at block 48,291,111, and the maker's sweep. All nine Zcash transactions of the two runs were fully shielded v6 Ironwood transactions with no transparent inputs or outputs.

10. Related Work

Hash-locked swaps. Nolan's construction [24], formalized by Herlihy [25], and XCAT for Zcash [26] need scripts with hash locks and timelocks.

Scriptless swaps. Gugger's Bitcoin–Monero protocol [27] locks the Monero leg to a split key and reveals a share through adaptor signatures, with timelocks t_0 and t_1 and a punish path; COMIT ran it in production from 2021 [28].

AthanorLabs replaced the Bitcoin leg with an Ethereum contract that checks a revealed secp256k1 scalar [29]. Its windows match those of our first design: claim when ready before its first timeout, or between the timeouts; refund before the first timeout unless ready, or after the second. Its claim can be submitted by a relayer for a fee, as in our design, so the claimer needs no ETH. Ed25519 shares require a cross-group equality proof [30], which a Pallas check avoids. Universal atomic swaps [31] need only signature verification on each chain. Han et al. identify the free option of atomic swaps [32].

Zcash. Zwap [34] offers Orchard–EVM swaps capped at $100; forum reviewers objected that its Orchard-side lock cannot be enforced, since Orchard has no scripts. A grant proposal for RedPallas adaptor signatures [35] closed without code. Zswap [33] is an unrelated scheme for multi-asset swaps within one shielded ledger.

Dark pools. Renegade [36] matches orders at the midpoint with MPC; Penumbra batches swaps and plans sealed-bid inputs [37]. Neither settles shielded ZEC.

11. Limitations

- **Assumptions.** The maker must satisfy L_M, and both parties must satisfy (H). Token issuers can freeze credits. A compromised device exposes the swaps it is running.
- **Sequencer failure.** We assume a live sequencer. If it censors or halts, the finalized head stalls, forced inclusion through Ethereum can take about 12 hours, and block timestamps can lag real time; neither D nor F is then bounded on the time scale of our windows. A batcher stall has a similar effect: the sequencer keeps producing blocks, but they stop becoming final, so every rule that waits for finality stalls. A deployment should stop opening swaps when the finalized head stalls. We have not modelled or tested forced inclusion. The budget R models a bounded number of unsafe-head reorganizations, such as follow from sequencer or batcher incidents; a sequencer that equivocates without bound is outside the model.
- **Privacy.** Timing is the binding constraint, and cover delays cost hours. Wallet fingerprints (fee, expiry) split the anonymity set. Each transaction's anchor, the commitment-tree root it proves against, is also public and wallet-specific; we did not measure anchor depths, so the full-settings class may be smaller than Table 7 estimates. The light wallet leaks its fetches to the server, and without Tor an external relayer learns the taker's network address. The maker is public.
- **Verification scope.** The TLA$^+$ model checks three small parameter points, all with $D = F$, and abstracts cryptography. Halmos proves bounded call sequences, and unrestricted three-call sequences did not terminate. The contracts have not had an external audit.
- **Implementation.** One maker per request; manual epoch-key rotation; no Tor; cover-delay waits block the maker's other swaps; our taker self-relays and does not implement a third-party relayer with fallback.
- **Not tested.** Forced inclusion, mainnet deployments, and production window lengths. Reorganizations were injected on the local EVM chain only.

12. Conclusion

Cloakroom settles trades between shielded ZEC and EVM tokens without custodians and without leaving the shielded pool. Its safety rests on proofs under explicit timing assumptions, and a model checker shows that each rule and each budget is needed. Review and model checking at a second parameter point found gaps in our first version, all fixed: the reorganization budget lacked one inclusion delay per reorganization; a dropped refund was not re-sent after its first deadline; the rule against setting Ready after publishing b was not stated; secrets were erased before the transactions that needed them were final; the escrow salt could be derived from a public identifier; and public EVM events anchored Zcash transactions from behind. Its privacy has been measured on mainnet rather than assumed. Shielding hides amounts and parties; timing is what remains, and how a cover delay is drawn matters as much as how long it is.

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Appendix: Test Vectors

$\mathcal{G}$ (from `orchard` 0.16.0):

```
x = 0x375523b328f1d6063b8d187c3e5f445f0c7f0ce37b70a10c8d1a7284b875c963
y = 0x1ad0357fdf1a66db7b10bcfcfed624fbdfc914fec005bdd84ce33e817b0c3bc9
```

A share a and $A = [a]\mathcal{G}$:

```
a   = 0x195661e2e4789b7f57137b2e777736d79720270638d12e4882fd95e52a73d6b3
A.x = 0x17a182b1eb9594581ecaad0fcb61546f4c366cc57b9575ddde22ca56963c304c
A.y = 0x19b252972f768cc5708b251baca77a9afd02dddd9531510aafb13c4824d7b61f
```